

Preterm Birth Prediction using GraphSAGE and Electrohysterogram Data

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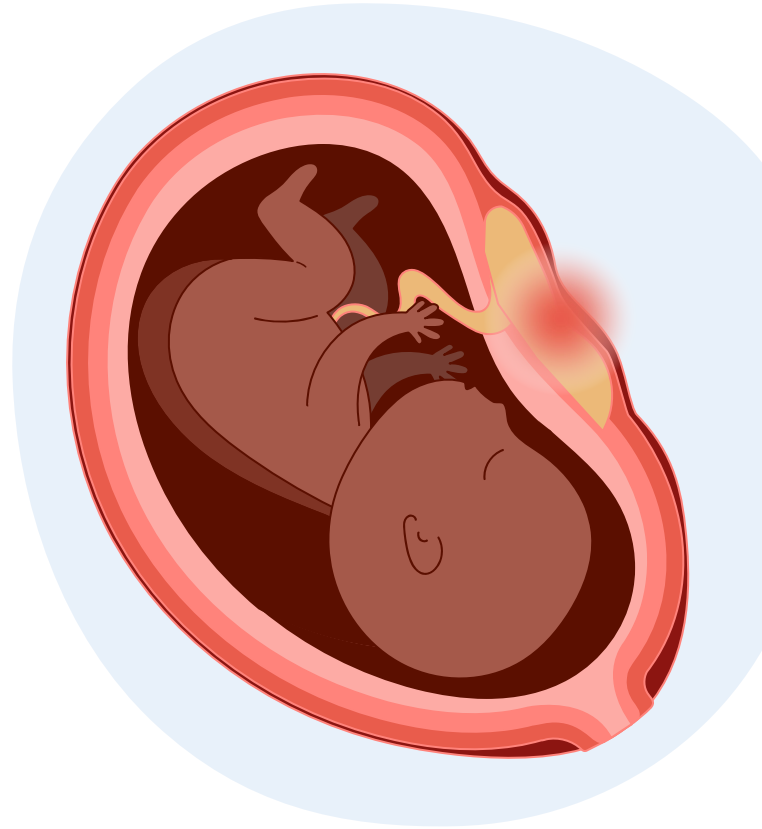


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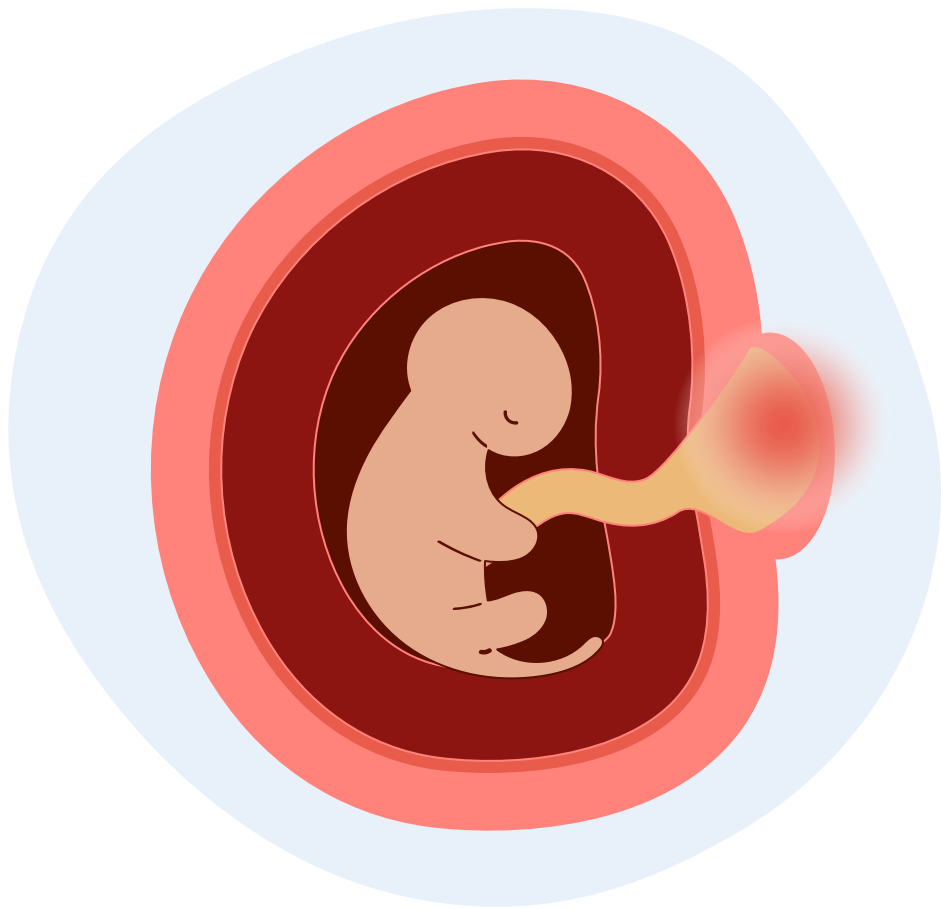
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01

Introduction

Introduction & Problem Statement

Preterm Birth: A Global Health Challenge

- Affects ~15 million infants annually worldwide
- Major cause of neonatal intensive care unit admissions 11.1% of fetuses born before due date

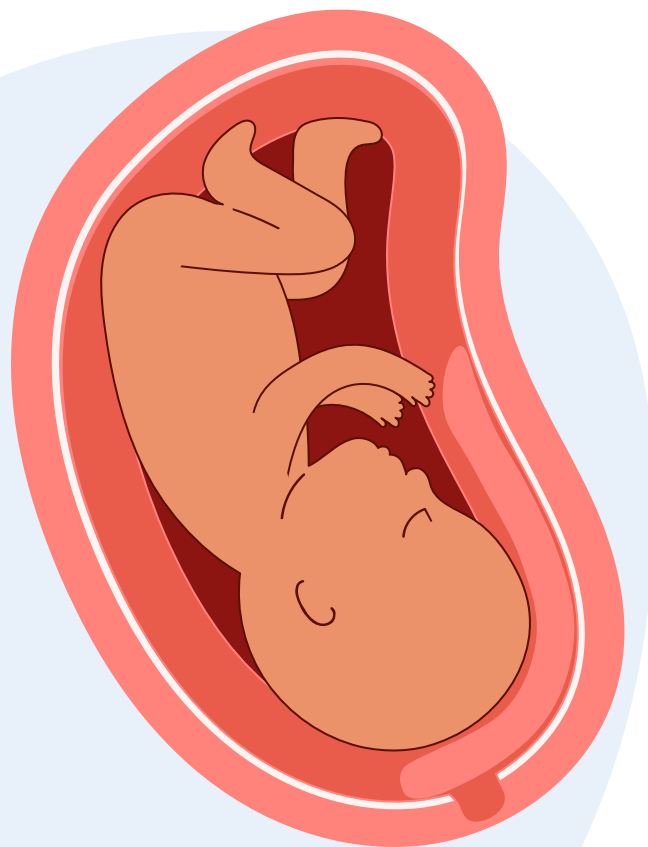
Electrohysterogram (EHG):

- Non-invasive method to monitor uterine electrical activity using surface electrodes.
- Helps detect myometrial contractions and predict preterm labor.

Our Approach

- We use Graph Neural Network (GraphSAGE) for EHG analysis with Explainable AI (SHAP).
- Ensures transparency and interpretability for clinical use.





02

Background and Literature review

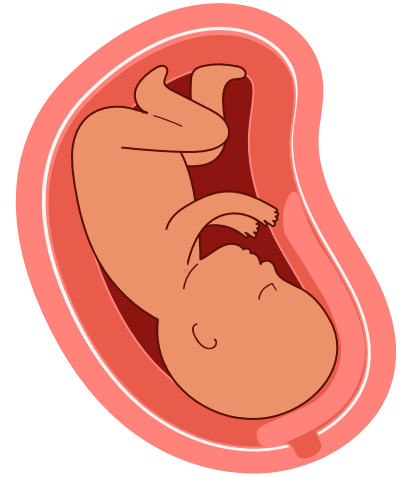
Background & Literature Review

Previous Approaches:

- Empirical mode decomposition with Random Forest: 99.23% accuracy.
- CNN MobileNet-V2 with Quadratic Discriminant classifier: 99.8% accuracy.
- Recurrent neural networks with short-time Fourier transform.
- Hybrid machine learning: Enhanced Sheep Flock Optimized Hybrid ANN
- Spiking Neural Network with Seagull Optimization Algorithm

Research Gap:

- Limited focus on model interpretability in clinical context
- Need for graph-based approaches to capture complex EHG signal relationships



03

Methodology



Dataset Overview

TPEHG DB Dataset:

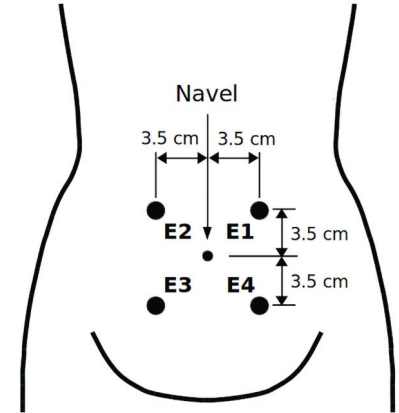
- 300 uterine EMG recordings (30 minutes each)
- Term deliveries: 262 records (>37 weeks)
- Preterm deliveries: 38 records (≤ 37 weeks) Gestation stages: 162 before 26th week, 138 after 26th week
- Sampling rate: 20 Hz per channel with 16-bit resolution

Recording Configuration:

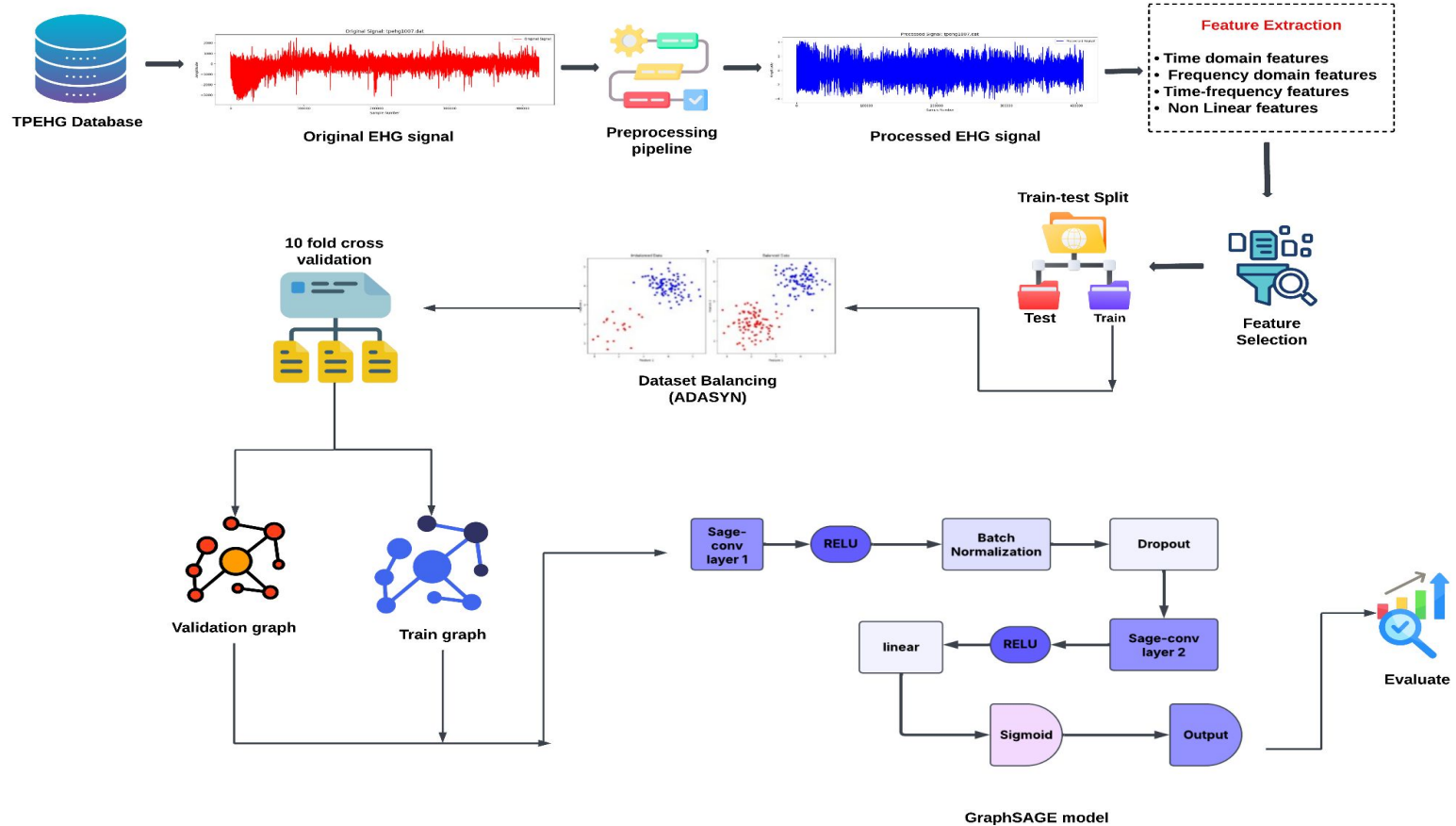
- 4 electrodes placed symmetrically around navel.
- 3 recording channels: S1 (E1-E2), S2 (E2-E3), S3 (E4-E3)

Validation Dataset:

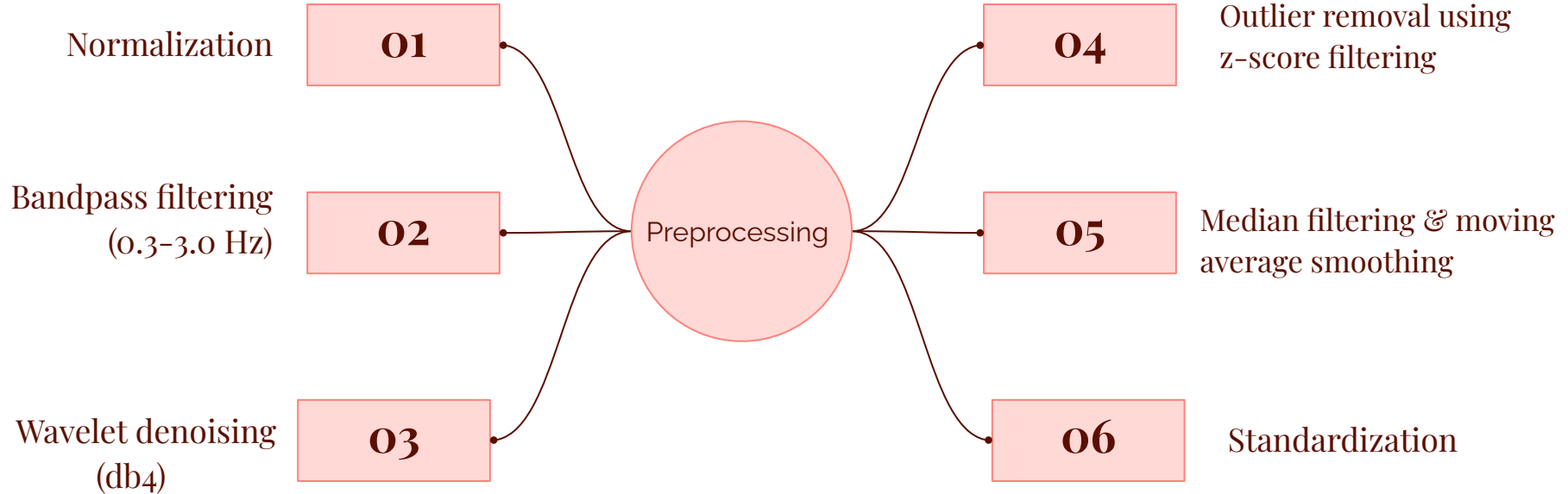
- TPEHGT 26 recordings: 13 preterm, 13 term.
- Same recording protocol as TPEHG



Block Diagram



Preprocessing pipeline





Feature Extraction

- **Time domain:** statistical features, IQR, RMS, zero crossings.
- **Frequency domain:** PSD, dominant frequency, spectral entropy.
- **Time-frequency:** wavelet entropy, wavelet energy (Level 1-5).
- **Nonlinear:** RQA features (recurrence rate, determinism, laminarity)
- **Clinical features** from dataset metadata



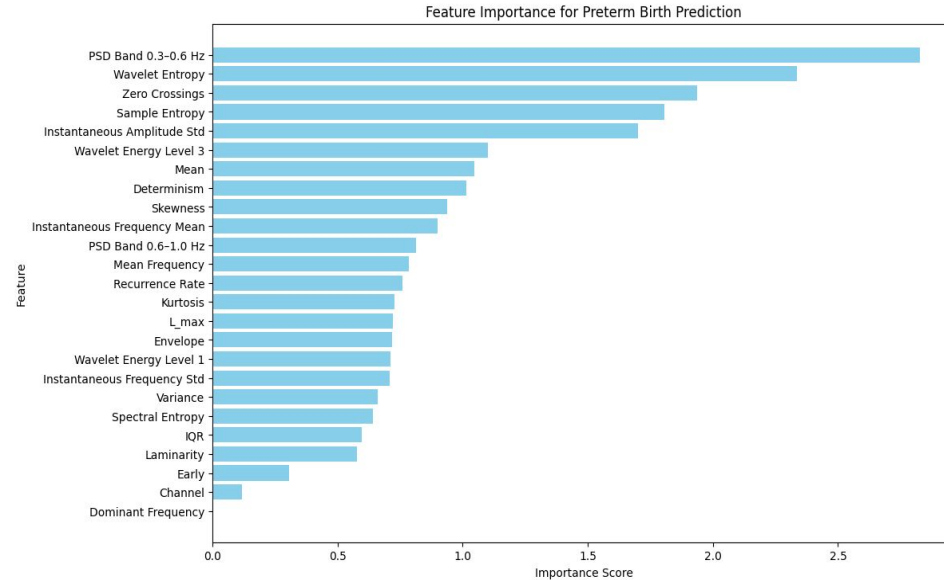
Data Balancing

- Applied to training data only (80/20 train-test split)
- Adaptive approach focusing on difficult-to-classify regions
- Generates synthetic samples for minority class (preterm)
- Advantages over SMOTE: reduced overfitting risk

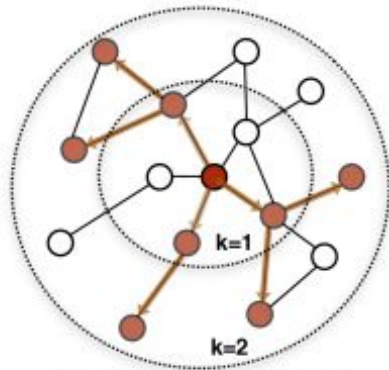


Feature Selection

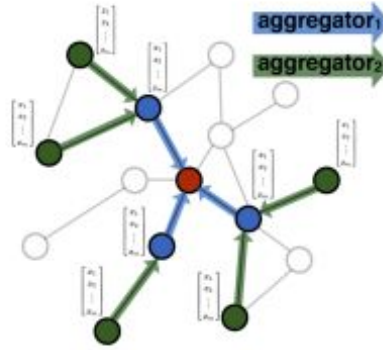
- Correlation-based filtering (threshold: 0.9)
- Ensemble feature selection:
 1. Random Forest importance
 2. Recursive Feature Elimination (RFE)
 3. L1 Regularization (Lasso)
- Min-Max scaling of importance scores
- Combined ranking for robust feature assessment



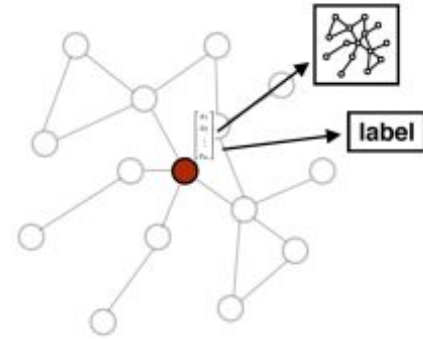
Working of GraphSAGE



1. Sample neighborhood



2. Aggregate feature information from neighbors



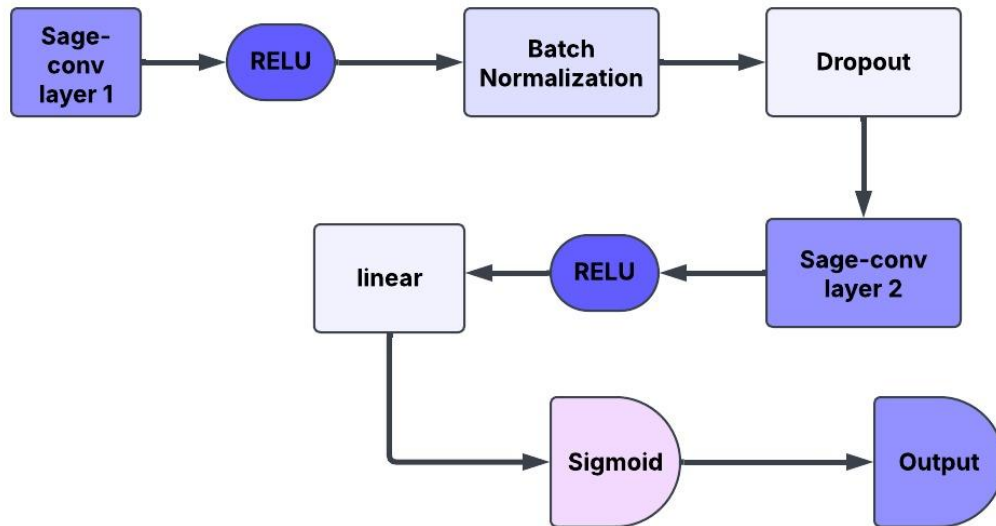
3. Predict graph context and label using aggregated information

Graph Construction:

- Patients as nodes with 38-dimensional feature vectors
- Edges based on cosine similarity (threshold: 0.85)
- Connects patients with similar clinical profiles

Model Architecture

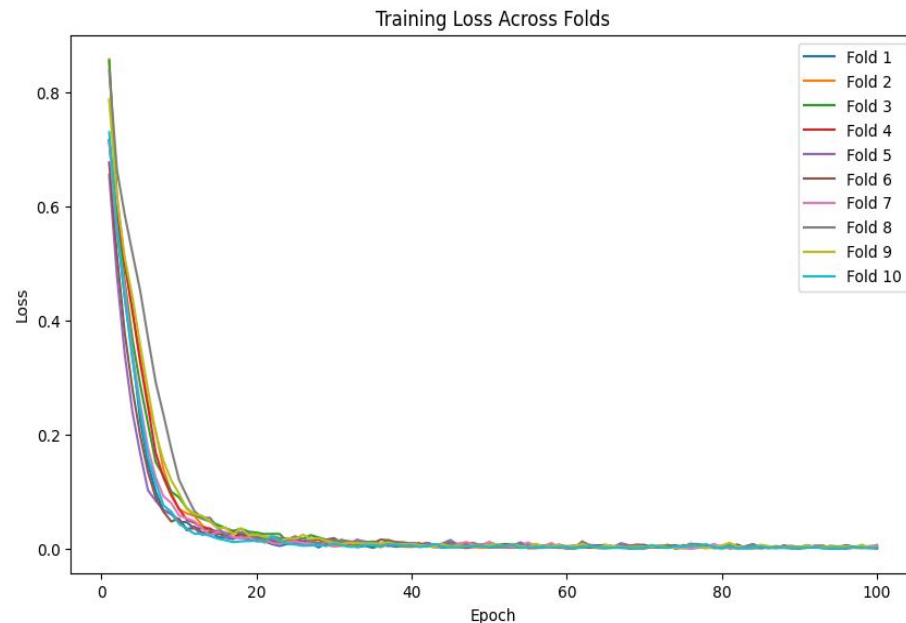
- **Input layer:** Standardized feature vectors
- Two SAGEConv layers with mean aggregation
- Batch normalization & dropout for stability
- Fully connected output layer for binary classification



Training - 10 fold cross validation

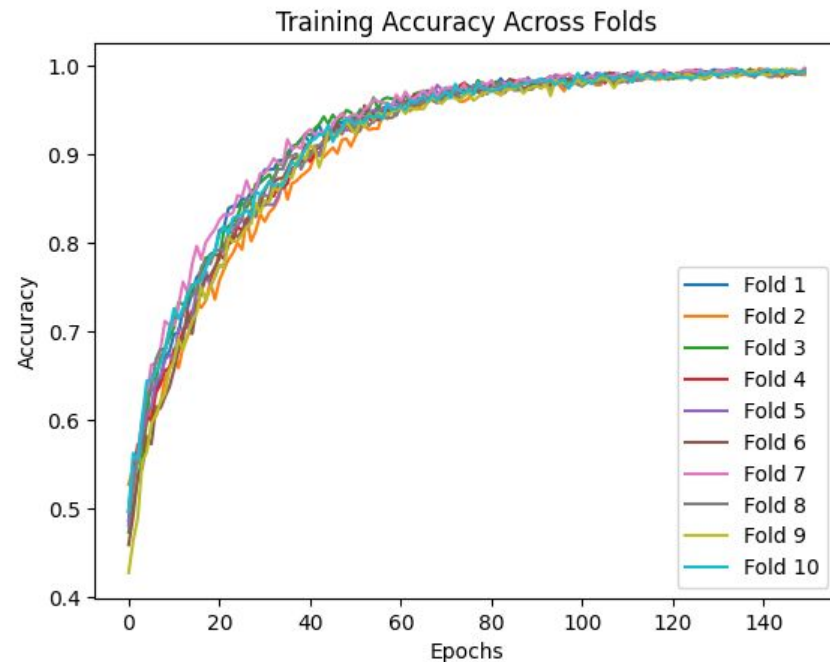
Training Methodology:

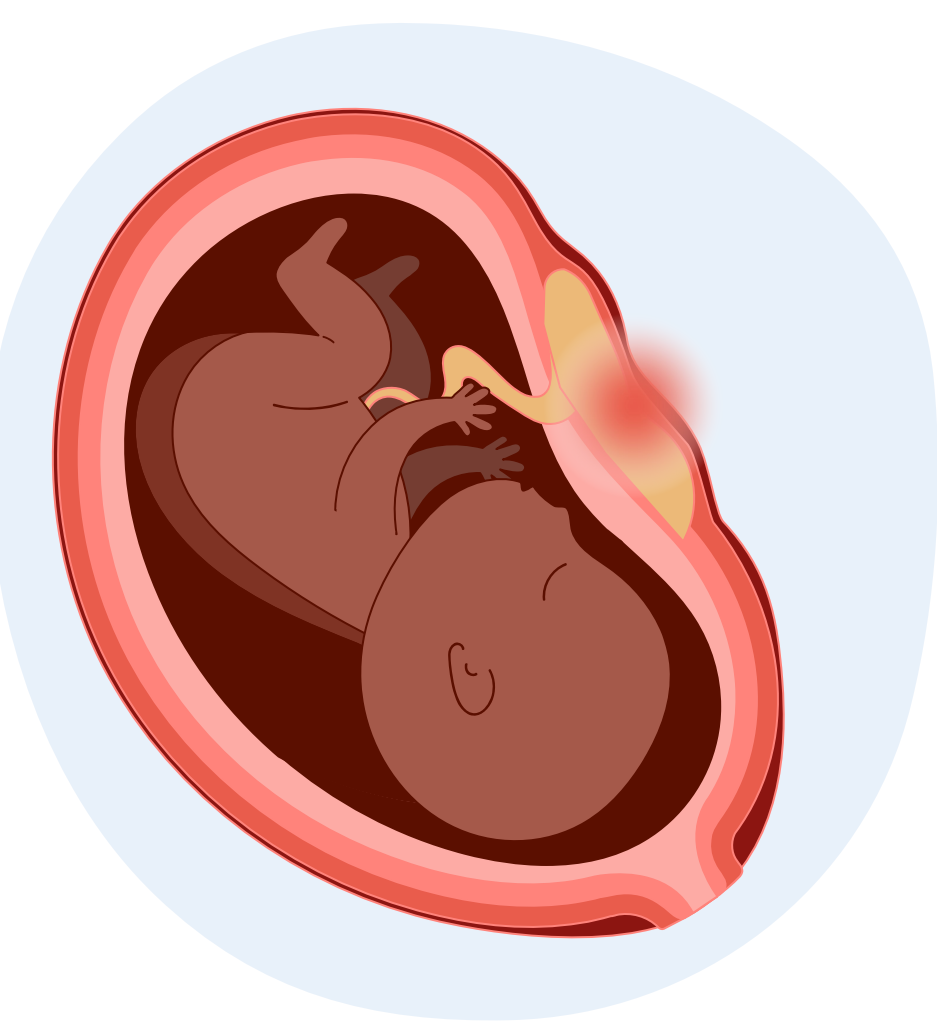
- Stratified 10-fold cross-validation
- 100 epochs per fold
- Adam optimizer (learning rate: 0.01, weight decay: 5×10^{-4})
- Tracking of training and validation accuracy, loss.



External Validation Approach

- Merged TPEHG with TPEHGT dataset
- Feature vector reduction (38 to 33 dimensions)
- Same preprocessing & feature selection pipeline
- Extended validation with 10-fold cross-validation

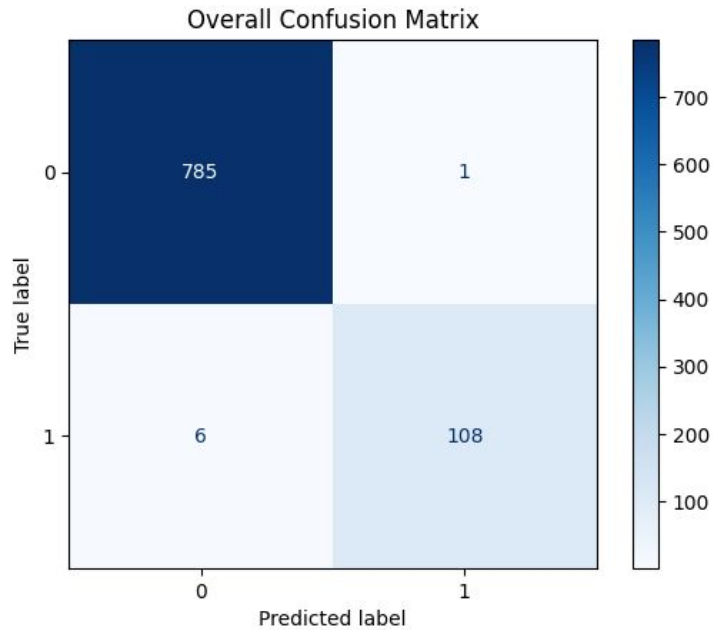




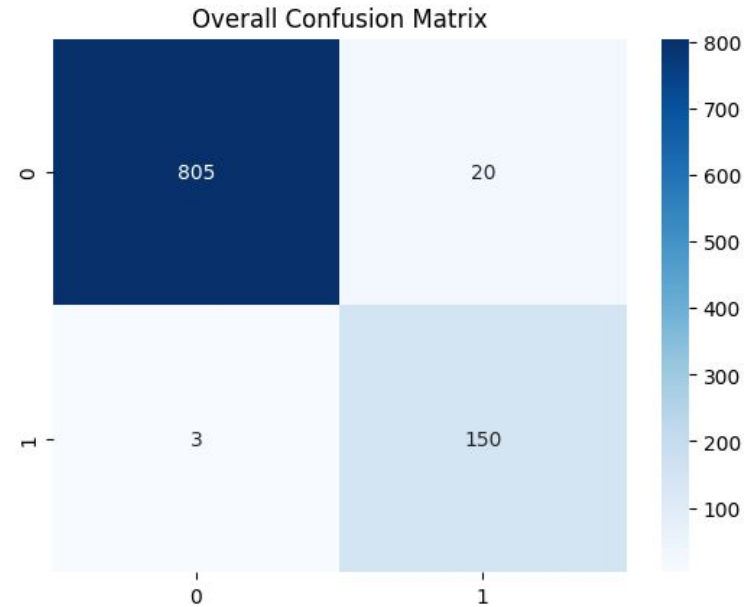
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Results

Results - Confusion Matrix



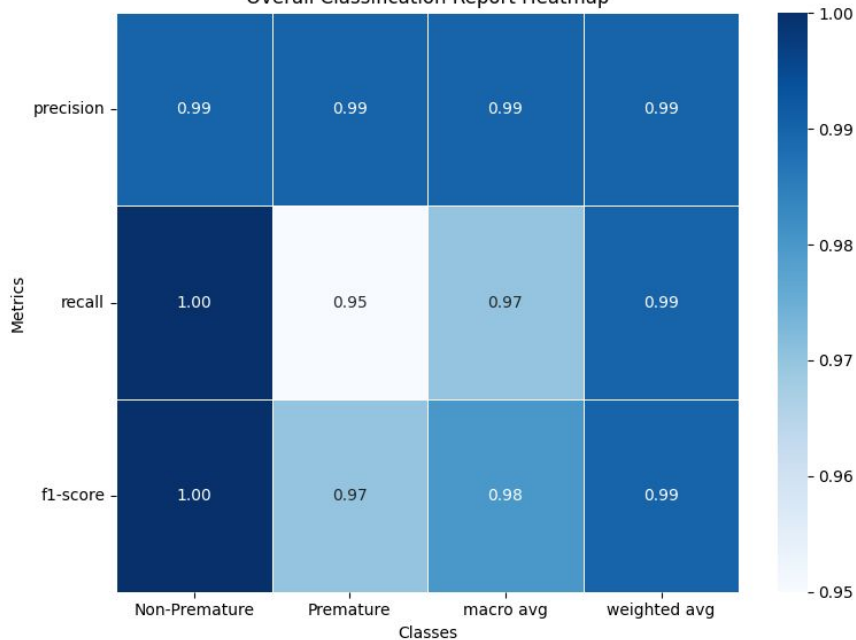
Only TPEHG



TPEHG + TPEHGT

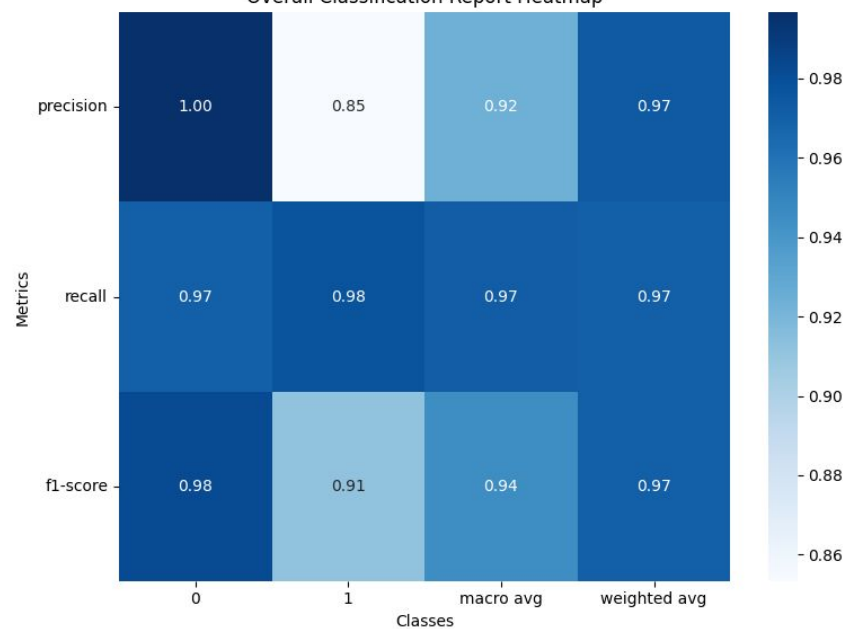
Results - Classification report heatmap

Overall Classification Report Heatmap



Only TPEHG

Overall Classification Report Heatmap



TPEHG + TPEHGT

Results- Comparison

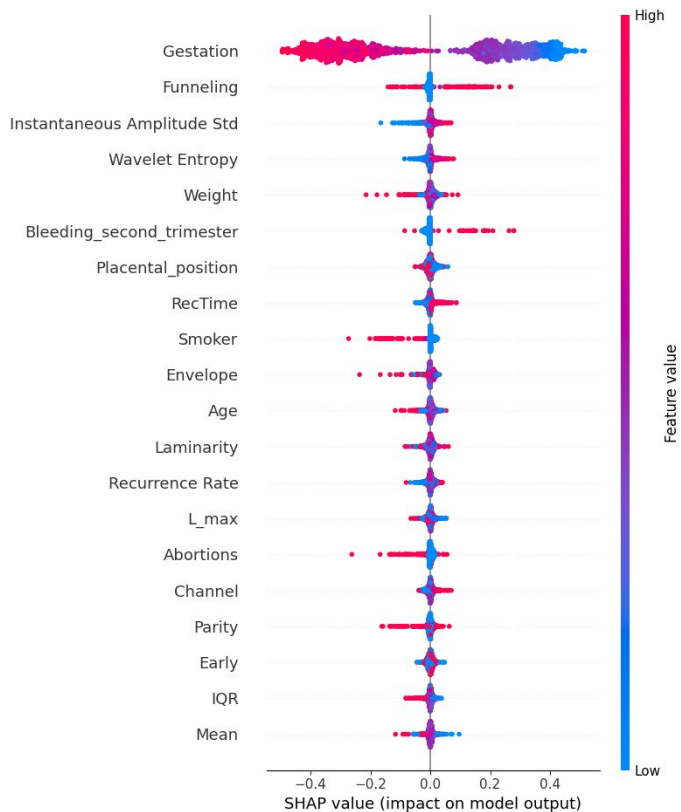
METRIC	TPEHG DATASET	MERGED DATASET
Overall Accuracy	99.22%	97.65%
Recall	94.74%	98.04%
Specificity	99.87%	97.58%
True Positives	108	150
True Negatives	785	805
False Positives	1	20
False Negatives	6	3
Key observations	- Higher accuracy and specificity but lower sensitivity. - Need more preterm samples for generalization	- Higher sensitivity indicates good generalization for preterm detection. - Missing clinical features may impact specificity.

05

SHAP Analysis



SHAP Summary Plot



Top influential features:

- Gestation period
- Cervical funneling
- Bleeding in second trimester
- Wavelet entropy
- Weight

SHAP Summary Plot Insights:

- Gestation values strongly impact preterm probability.
- Presence of funneling increases preterm probability
- Bleeding history significantly impacts prediction.

Conclusion and Future Work



Key Contributions

- First application of GraphSAGE for PTB prediction from EHG.
- 99% accuracy on TPEHG dataset (10-fold CV).
- Strong generalization (97.65% accuracy on merged dataset)
- Explainable predictions through SHAP analysis
- Identification of critical clinical and signal features



Future Work

- Testing with additional EHG datasets
- Exploring deeper graph-based architectures
- Enhancing interpretability mechanisms
- Real-time monitoring system development Clinical validation studies

Thank You